

Airline On-Time Performance

Final Data Analysis

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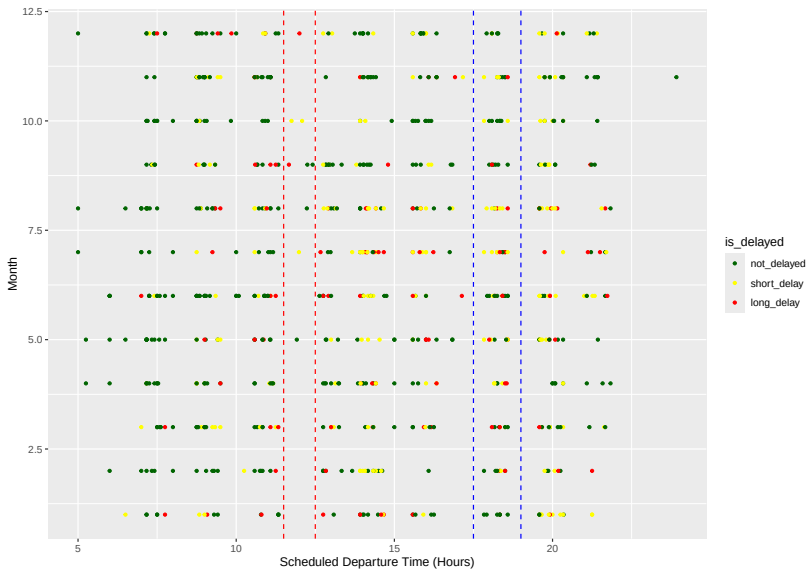
Review of scope

- ▶ Identifying flights at high risk of delay months in advance is a key challenge for United Airlines' network planning team
- ▶ Models will be built using historical flight data to predict the probability of departure delays at ORD (Chicago O'Hare International Airport) and PIT (Pittsburgh International Airport)
- ▶ Ideally, these models will allow planners to estimate operational impact to make better decisions
- ▶ For the scope of this project, we'll take United Airlines data between October 2024 and September 2025

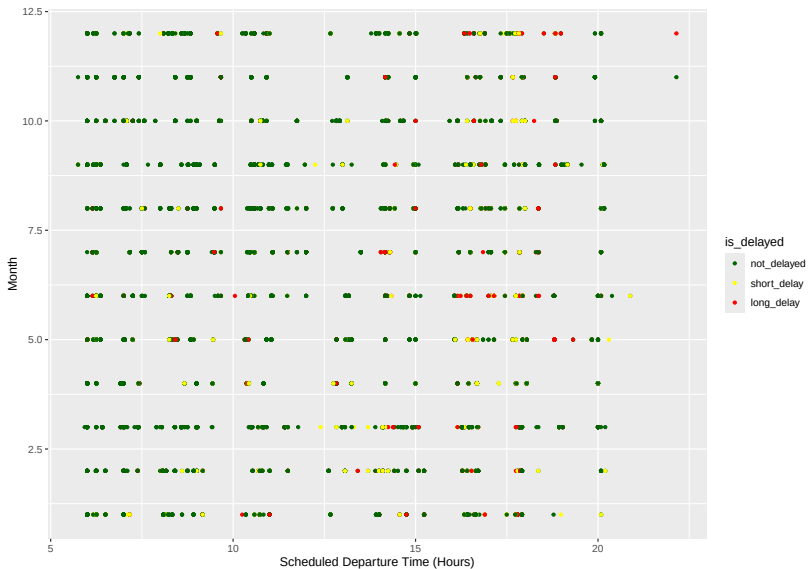
Bank Background

- ▶ “Banks” are a schedule structure tool that allow airlines to provide better connectivity across a network
- ▶ Between the red lines is an example of an arrival bank (few departures)
- ▶ Between the blue lines is an example of a departure bank (many departures)
- ▶ These alternate throughout the day at hub airports

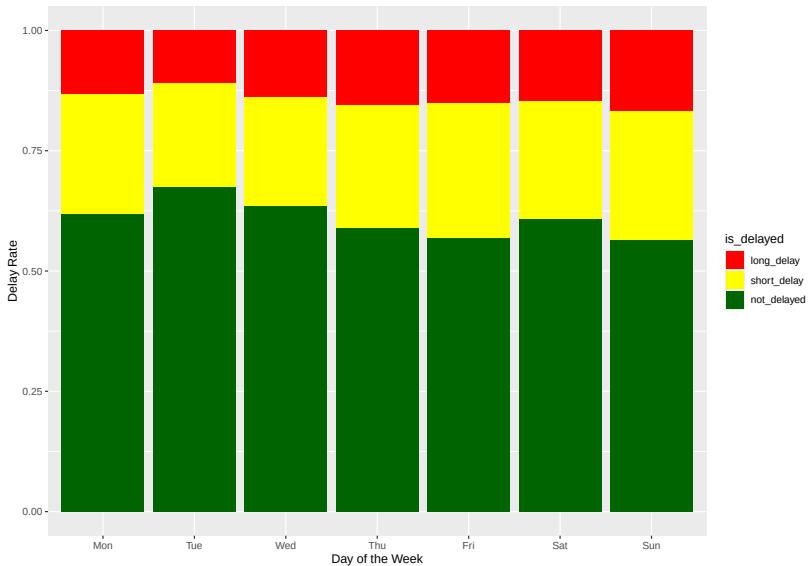
Bank Structure at ORD



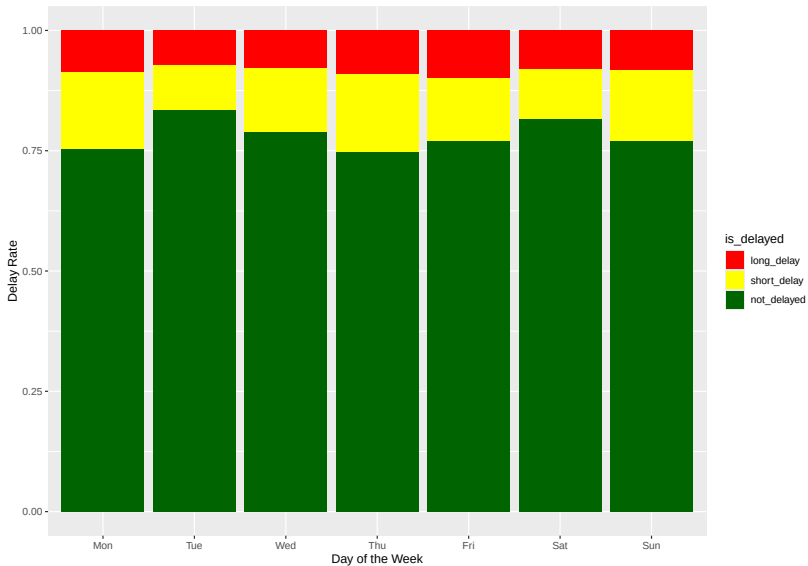
Lack of Bank Structure at PIT



Delay rate by Day of Week at ORD



Delay rate by Day of Week at PIT

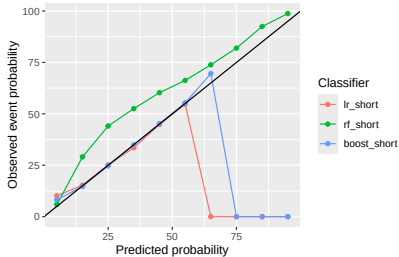


Model Fitting

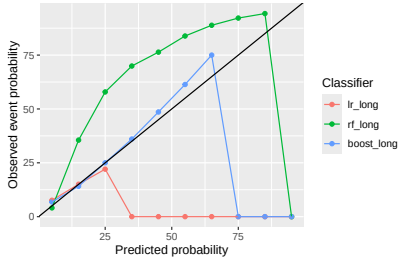
- ▶ ORD Breakdown: On-Time = 60.8%, Short Delay = 24.9%, Long Delay = 14.3%
- ▶ PIT Breakdown: On-Time = 78.2%, Short Delay = 13.3%, Long Delay = 8.5%
- ▶ Predictors include Scheduled Departure Time, Scheduled Arrival Time, Scheduled Block Time, Day of the Week, Month, and Scheduled flights in 30 minutes before scheduled departure time (NEW)
- ▶ Three models will be fit for ORD and PIT: Logistic Regression, Random Forest, and Boosting
- ▶ Models will be evaluated on calibration and bias

Model Calibration

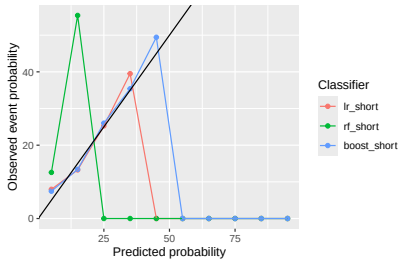
ORD – Short Delay Calibration



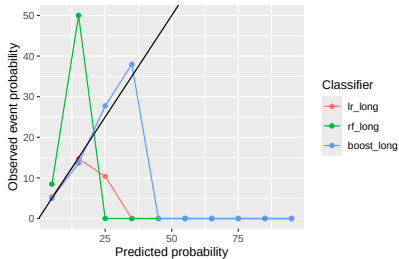
ORD – Long Delay Calibration



PIT – Short Delay Calibration



PIT – Long Delay Calibration



Model Calibration Continued

Table 1: ORD - Short Delay Calibration Summary

Bin	Logistic Regression		Random Forest		Boosting	
	Pct	N	Pct	N	Pct	N
[0,0.1]	10.166	227	5.898	5593	8.259	646
(0.1,0.2]	15.333	9789	29.043	8252	14.672	8454
(0.2,0.3]	25.154	13465	44.081	7772	24.679	14568
(0.3,0.4]	33.504	14634	52.546	6273	34.890	11694
(0.4,0.5]	44.798	5171	60.261	5174	45.270	6374
(0.5,0.6]	54.712	447	66.245	3774	55.326	1901
(0.6,0.7]	0.000	0	73.823	2995	69.565	96
(0.7,0.8]	0.000	0	81.970	2155	0.000	0
(0.8,0.9]	0.000	0	92.477	1426	0.000	0
(0.9,1]	0.000	0	98.762	319	0.000	0

Model Calibration Continued

Table 2: ORD - Long Delay Calibration Summary

Bin	Logistic Regression		Random Forest		Boosting	
	Pct	N	Pct	N	Pct	N
[0,0.1]	7.600	3659	4.064	5572	6.803	3781
(0.1,0.2]	15.054	14162	35.542	6952	14.112	12713
(0.2,0.3]	22.057	7371	57.892	5384	25.003	5367
(0.3,0.4]	0.000	0	69.930	3514	36.016	2367
(0.4,0.5]	0.000	0	76.378	1940	48.592	880
(0.5,0.6]	0.000	0	83.866	1050	61.364	81
(0.6,0.7]	0.000	0	88.854	558	75.000	3
(0.7,0.8]	0.000	0	92.195	189	0.000	0
(0.8,0.9]	0.000	0	94.286	33	0.000	0
(0.9,1]	0.000	0	0.000	0	0.000	0

Model Calibration Continued

Table 3: PIT - Short Delay Calibration Summary

Bin	Logistic Regression		Random Forest		Boosting	
	Pct	N	Pct	N	Pct	N
[0,0.1]	7.965	229	12.572	981	7.453	244
(0.1,0.2]	13.252	533	55.385	72	13.406	470
(0.2,0.3]	25.271	210	0.000	0	26.007	226
(0.3,0.4]	39.512	81	0.000	0	35.385	69
(0.4,0.5]	0.000	0	0.000	0	49.438	44
(0.5,0.6]	0.000	0	0.000	0	0.000	0
(0.6,0.7]	0.000	0	0.000	0	0.000	0
(0.7,0.8]	0.000	0	0.000	0	0.000	0
(0.8,0.9]	0.000	0	0.000	0	0.000	0
(0.9,1]	0.000	0	0.000	0	0.000	0

Model Calibration Continued

Table 4: PIT - Long Delay Calibration Summary

Bin	Logistic Regression		Random Forest		Boosting	
	Pct	N	Pct	N	Pct	N
[0,0.1]	5.296	276	8.475	672	4.996	277
(0.1,0.2]	14.715	392	50.000	2	13.646	265
(0.2,0.3]	10.345	6	0.000	0	27.717	102
(0.3,0.4]	0.000	0	0.000	0	37.975	30
(0.4,0.5]	0.000	0	0.000	0	0.000	0
(0.5,0.6]	0.000	0	0.000	0	0.000	0
(0.6,0.7]	0.000	0	0.000	0	0.000	0
(0.7,0.8]	0.000	0	0.000	0	0.000	0
(0.8,0.9]	0.000	0	0.000	0	0.000	0
(0.9,1]	0.000	0	0.000	0	0.000	0

Model Application Scenario

- ▶ You work on the network planning team and a complaint has come from the operations analytics team about an under performing flight
- ▶ Your goal is to find a departure time that reduces expected delay rate
 - ▶ In a real scenario decision should balance long delay rate, short delay rate, and pricing info

Find Underperforming Flight

Flt Num	Arr Airport	Date	Day	Sch Dep Time	Delay Min
312	CHS	2025-07-02	Wed	18:15	21
312	CHS	2025-07-09	Wed	18:15	-4
312	CHS	2025-07-16	Wed	18:15	382
312	CHS	2025-07-23	Wed	18:15	74
312	CHS	2025-07-30	Wed	18:15	22

- ▶ Predict probabilities for short and long delays every 5 minutes and compare expected delay rates
 - ▶ Block time = 138 min (generally doesn't change much year to year)
 - ▶ 30 minute before aggregate was taken from 07/09/2025 (schedule doesn't change much year to year)

Apply Model

